

Laboratory Report: Logical Reasoning for Planning

Course: CSF407 / Artificial Intelligence Laboratory | Date: September 30, 2026 | Topic: STRIPS Planning, Search & Prolog Verification

1. Task 0: Understand the Planning Problem

A planning problem is represented by the 3-tuple $\Pi = (I, A, G)$:

- Initial state I:** $\{At(Robot, A), At(Package, A)\}$ | **Goal G:** $\{At(Package, C)\}$.
- Actions:** $Move(X, Y)$, $PickUp(Package, X)$, $Drop(Package, X)$ with positive/negative preconditions and add/delete effects.
- Applicability:** $PickUp(Package, A)$ is **applicable** in I because all its preconditions are present. $Drop(Package, C)$ is **inapplicable** because the robot is neither at C nor holding the package.

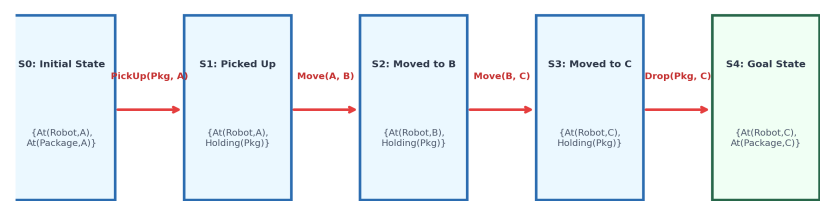
Think About It: An action is applicable if and only if $S \models Preconditions(a)$. Logic provides the mathematical bridge determining whether an action can physically take place.

2. Task 1: Manual Plan Construction

Optimal 4-step sequence: $S_0 \rightarrow Pickup(P, A) \rightarrow Move(A, B) \rightarrow Move(B, C) \rightarrow Drop(P, C) \rightarrow S_4$.

- $S_0: \{At(Robot, A), At(Package, A)\}$
- $S_1: \{At(Robot, A), Holding(Package)\}$
- $S_2: \{At(Robot, B), Holding(Package)\}$
- $S_3: \{At(Robot, C), Holding(Package)\}$
- $S_4: \{At(Robot, C), At(Package, C)\} \models G$.

Task 1: Valid State Transition Sequence for Warehouse Goal Achievement



3. Task 2 & 3: Prompt Engineering and Systematic Testing

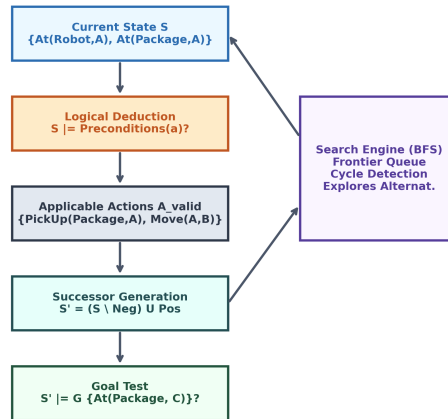
Prompt: 'Implement planning agent in Python. Represent state as set of propositions. Action contains pos/neg preconds and pos/neg effects. Applicability: all preconds satisfied. Apply: remove neg effects, add pos effects. Use BFS to find plan achieving goal. Detect when no plan exists.'

Test Case	Condition / Modification	Expected Result	Observed Result	Status
Test A: Solvable	Standard warehouse problem	4-action plan	4 actions, 6 expanded	PASS
Test B: Impossible	Removed Pickup action	Reports 'No plan found'	None returned, 3 expanded	PASS
Test C: Irrelevant	Robot moves without package	Strictly achieves At(Pkg,C)	Ignored empty moves	PASS

4. Task 4: Logic + Search = Planning

- Logical Reasoning:** Operates locally on states and actions, evaluating entailment $S \models Preconditions(a)$ and computing successor states $S' = (S \setminus Neg) \cup Pos$. Logic determines **what is possible**.
- State-Space Search (BFS):** Operates globally, exploring candidate action sequences to reach the goal. Search determines **what to try**.

Task 4: Logical Reasoning + State-Space Search = Planning



5. Task 5: LLM Explanation vs Independent Verification

An LLM produces persuasive textual explanations, but lacks an internal proof engine and may hallucinate satisfied preconditions.

Independent state transitions must always be trusted over LLM explanations. A generated explanation is not an independent verification.

6. Tasks 6, 7 & 8: Prolog as an Independent Rule-Based Verifier

In planner.pl, the warehouse topology is encoded as Horn clauses:

- **Facts:** connected(a,b). connected(b,a). connected(b,c). connected(c,b).
- **Rules:** can_move(X,Y) :- connected(X,Y). valid_move(X,Y) :- connected(X,Y).
- **Queries:** ?- valid_move(a,b) → **true**. | ?- valid_move(a,c) → **false**. (Catches invalid direct leap!)
- **Task 8 Deduction Chain:** wet_road ⇒ (wet_road ⇒ slippery) ⇒ (slippery ⇒ reduce_speed) ⇒ **reduce_speed**.

7. Key Reflection Answers

- **Preconditions/Effects:** Prevent the LLM from inventing illegal transitions or teleporting packages.
- **Unchecked Preconditions:** Would allow the robot to drop the package at C without picking it up or traveling there.
- **Logical Reasoning Role:** Evaluates precondition satisfaction, generates state mutations, and runs Prolog inference.
- **Relation to Search:** Planning searches a graph where states are sets of logical propositions and transitions are logical deduction operators.